RV College of Engineering®
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Academic year 2024-2025 (Odd Sem)
DEPARTMENT OF

COMPUTER SCIENCE & ENGINEERING

Date	5th Feb-25	Maximum Marks	10+50
Course Code	CS354TA	Duration	120 Min
Sem-V	ReTest	Staff: HKK|ASP|SMS|MRA	

THEORY OF COMPUTATION
(Common to CSE & ISE)

PART-A

		Marks	BT	CO
1.	Consider the regular expressions, r=0*+1*, s=01*+10*+1*0+(0*1)*, t=1*(01)*0* a) Find a string of minimum length (excluding empty string) not in the language corresponding to the regular expression t. 001 b) Find a string corresponding to s but not in t. 011	2	2	1
2.	Construct Ɛ-NFA for the regular expressions *(b + bba)*ab**, so that there is a recognizable correspondence between the regular expression and the transition diagram.	2	2	1
3.	Prove that the language L = {vwv: v, w ∈{a, b}*, |v| =2} is regular.	1	3	2
4.	Construct DPDA to accept the language which is equivalent to the language generated by the regular expression (a+b)*abb.	2	2	2
5.	Describe the language generated by the right linear grammar productions A-->aB|bC|a, B-->aC|bA, C-->aA|bB|b	1	1	2
6.	For the right linear grammar with productions S→aA|bC|b, A→aS|bB, B→aC|bA|a, C→aB|bS. Find the equivalent left linear grammar.	1	2	3
7.	Let G be the CFG with productions {S-->aB|bA, A-->a|aS|bAA, B-->b|bS|aBB}. How many steps are there to derive a string w ϵ L(G) where |w|=n.	1	2	2

PART-B

1a.	Construct DFA for L_1-L_2 where L_1={w| n_a(w) mod 3=1} and L_2={w| n_b(w)mod 2 =0}	5	3	2
1b.	Minimize the states of the following DFA by table filling algorithm.	5	3	3

δ	0	1
A	B	E
B	C	F
*C	D	H
D	E	H
E	F	I
*F	G	B
G	H	B
H	I	C
*I	A	E

2.	Construct ε NFA to accept any number of a's followed by any number of b's followed by any number of c's. Also convert to DFA by first converting to NFA and then to DFA.	10	4	3
3a.	Define CFG and the language generated by CFG. Give CFG for the following languages. i. Language corresponds to (abb+b)*(ab)*. ii. L= {w | w∈ {a, b}* and Na (w) ≠ N_b (w)}.	6	3	3
3b.	Decide in each case whether the given language is a CFL, and prove your answer. i. L={$a^n b^m a^m b^n$ |m, n ≥0 }. ii. L={xcx |x ∈ {a, b}*}	4	2	1
4.	In each case, given the context free grammar G, find an equivalent simplified CFG. i. S→ABC | BaB, A→aA | BaC | aaa, B→bBb | a, C→CA | AC ii. S→AB | AC, A→aAb | bAa | a | ϵ, B→bbA | aaB | AB, C→abCa | aDb | ϵ, D→bD | aC	10	2	1
5.	Define GNF grammar. Given below a CFG G, find a CFG G' in GNF generating L(G) – { ϵ }. A→BC, B→CA | b, C→AB | a	10	4	3